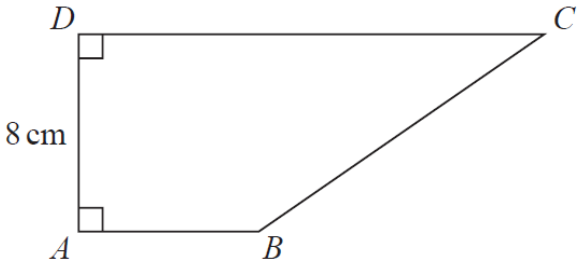
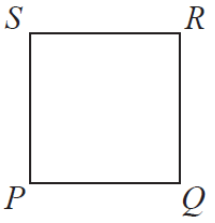
7 The diagram shows a trapezium $ABCD$ and a square $PQRS$.





$AD = 8$ cm
$DC = 3AB$

The perimeter of the square is 24 cm.

The area of the square is half the area of the trapezium.

Work out the length of AB.

Square side = 24 ÷ 4 = 6 cm

1 mark
Find square area.

Square area = 6^2 = 36 cm^2

Trapezium area = 2 × 36 = 72 cm^2

Let AB = t, so DC = 3t.

1 mark
Use trapezium formula.

A = ½(a + b)h: 72 = ½(t + 3t) × 8

1 mark
Set equal to 72.

72 = 16t, so AB = 72 ÷ 16 = 4.5 cm

1 mark
AB = 4.5 cm.

..... cm

(Total for Question 7 is 4 marks)